

## ERZURUM BOLGE, Turkey

WMO: 170950

Lat: 39.9058N

Lon: 41.2544E

Elev: 1860

StdP: 80.89

Time Zone: 3.00 (E03)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.1      | -22.1 | -28.1                         | 0.4 | -24.5 | -25.0 | 0.5 | -21.9 | 13.7                  | -3.0 | 10.9 | -5.0 | 0.2                   | 60   | 0.568 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 15.8                   | 31.0            | 14.0 | 29.7 | 13.9 | 28.3 | 14.0 | 17.1                | 26.6 | 16.3 | 25.6 | 15.3 | 24.7 | 3.0                  | 260  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.4                            | 12.0 | 22.1 | 12.4 | 11.3 | 21.1 | 11.6 | 10.7 | 20.2 | 55.8          | 26.4 | 52.8 | 25.8 | 49.8 | 24.4 | 24.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 10.2              | 8.9  | 7.1 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 6.4   | -8.6  | -6.3  | -0.9  | 6.1   | 10.8  | 15.3  | 20.0  | 20.6  | 15.0  | 8.8   | 0.1   | -4.9 | (6)  |
| (7)  |                                                                                  | DBStd       | 10.82 | 6.06  | 5.74  | 5.83  | 3.80  | 2.83  | 3.06  | 2.50  | 2.59  | 3.46  | 3.25  | 5.09  | 6.13 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2343  | 576   | 456   | 338   | 123   | 25    | 1     | 0     | 0     | 4     | 62    | 296   | 462  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4512  | 834   | 689   | 596   | 367   | 234   | 99    | 12    | 11    | 107   | 297   | 546   | 720  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1033  | 0     | 0     | 0     | 6     | 49    | 160   | 311   | 328   | 156   | 23    | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 162   | 0     | 0     | 0     | 0     | 0     | 8     | 64    | 81    | 9     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 2343  | 0     | 0     | 0     | 0     | 8     | 157   | 890   | 1064  | 224   | 1     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 706   | 0     | 0     | 0     | 0     | 0     | 19    | 280   | 369   | 38    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 1.9   | 1.5   | 1.3   | 1.9   | 2.3   | 2.3   | 2.1   | 2.3   | 2.1   | 1.8   | 1.9   | 1.3   | 1.3  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 467   | 29    | 37    | 51    | 70    | 67    | 40    | 21    | 15    | 21    | 45    | 37    | 35   | (15) |
| (16) |                                                                                  | PrecMax     | 556   | 84    | 112   | 93    | 166   | 119   | 78    | 47    | 50    | 66    | 95    | 77    | 77   | (16) |
| (17) |                                                                                  | PrecMin     | 327   | 4     | 18    | 8     | 25    | 28    | 7     | 2     | 1     | 0     | 5     | 1     | 4    | (17) |
| (18) |                                                                                  | PrecStd     | 65    | 15    | 19    | 21    | 28    | 25    | 20    | 11    | 11    | 16    | 25    | 21    | 19   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 3.5   | 6.4   | 13.5  | 19.8  | 24.3  | 28.6  | 32.8  | 32.6  | 29.1  | 22.3  | 13.2  | 6.4  | (19) |
| (20) |                                                                                  |             | MCWB  | 0.2   | 2.6   | 5.6   | 11.4  | 11.9  | 14.7  | 14.7  | 13.2  | 11.5  | 9.9   | 4.9   | 2.4  | (20) |
| (21) |                                                                                  | 2%          | DB    | 2.1   | 4.3   | 10.3  | 17.2  | 21.6  | 26.9  | 31.0  | 31.2  | 27.9  | 20.1  | 11.4  | 4.3  | (21) |
| (22) |                                                                                  |             | MCWB  | -0.6  | 1.2   | 4.2   | 8.7   | 11.5  | 14.4  | 14.7  | 13.5  | 12.0  | 9.3   | 4.5   | 1.7  | (22) |
| (23) |                                                                                  | 5%          | DB    | 0.8   | 2.9   | 8.5   | 15.4  | 19.6  | 25.0  | 29.6  | 30.1  | 26.1  | 17.8  | 9.3   | 3.2  | (23) |
| (24) |                                                                                  |             | MCWB  | -1.3  | 0.5   | 3.4   | 7.7   | 10.7  | 13.5  | 14.7  | 13.5  | 11.9  | 9.1   | 3.6   | 1.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | -0.2  | 1.4   | 6.4   | 13.4  | 17.9  | 23.2  | 28.1  | 28.9  | 24.1  | 15.5  | 7.2   | 2.1  | (25) |
| (26) |                                                                                  |             | MCWB  | -2.1  | -0.4  | 2.6   | 6.9   | 10.1  | 12.8  | 15.0  | 13.8  | 11.7  | 8.1   | 2.8   | 0.4  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 0.7   | 3.0   | 6.8   | 12.1  | 14.3  | 17.1  | 19.3  | 17.2  | 15.3  | 12.0  | 6.7   | 3.3  | (27) |
| (28) |                                                                                  |             | MCDWB | 2.5   | 6.0   | 10.7  | 17.8  | 21.8  | 25.7  | 28.9  | 27.3  | 22.5  | 19.4  | 10.0  | 5.2  | (28) |
| (29) |                                                                                  | 2%          | WB    | -0.2  | 1.6   | 5.0   | 10.1  | 12.5  | 15.5  | 17.8  | 16.2  | 13.7  | 10.6  | 5.5   | 2.2  | (29) |
| (30) |                                                                                  |             | MCDWB | 1.7   | 3.6   | 10.0  | 15.5  | 19.6  | 24.3  | 27.3  | 26.4  | 23.9  | 17.7  | 9.8   | 3.8  | (30) |
| (31) |                                                                                  | 5%          | WB    | -1.1  | 0.6   | 3.7   | 8.8   | 11.4  | 14.6  | 16.9  | 15.4  | 12.7  | 9.6   | 4.3   | 1.4  | (31) |
| (32) |                                                                                  |             | MCDWB | 0.8   | 2.4   | 7.5   | 13.7  | 18.0  | 23.1  | 26.3  | 25.9  | 23.4  | 16.0  | 8.5   | 2.8  | (32) |
| (33) |                                                                                  | 10%         | WB    | -2.2  | -0.3  | 2.7   | 7.5   | 10.4  | 13.7  | 16.0  | 14.8  | 12.0  | 8.7   | 3.1   | 0.5  | (33) |
| (34) |                                                                                  |             | MCDWB | -0.4  | 1.4   | 6.3   | 12.4  | 16.5  | 21.9  | 25.0  | 25.6  | 22.3  | 14.3  | 6.6   | 1.9  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 9.5   | 10.1  | 10.0  | 11.6  | 12.1  | 13.9  | 15.7  | 15.8  | 16.0  | 12.0  | 10.2  | 8.2   | (35) |      |
| (36) |                                                                                  | 5% DB       | MCDBR | 7.4   | 8.7   | 11.3  | 14.9  | 15.8  | 16.3  | 17.4  | 17.2  | 18.5  | 16.0  | 12.8  | 6.7  | (36) |
| (37) |                                                                                  |             | MCWBR | 5.4   | 6.2   | 6.8   | 8.7   | 8.2   | 7.0   | 6.8   | 5.8   | 7.6   | 8.0   | 7.3   | 4.6  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 6.8   | 8.2   | 10.3  | 14.1  | 14.5  | 15.7  | 16.5  | 16.2  | 18.1  | 14.1  | 11.1  | 5.7  | (38) |
| (39) |                                                                                  |             | MCWBR | 5.2   | 6.2   | 6.6   | 10.1  | 8.5   | 8.2   | 8.0   | 6.7   | 8.9   | 7.8   | 6.6   | 4.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.269 | 0.306 | 0.359 | 0.416 | 0.385 | 0.349 | 0.363 | 0.367 | 0.317 | 0.344 | 0.292 | 0.265 | (40) |      |
| (41) |                                                                                  | taud        | 2.242 | 2.116 | 2.063 | 2.031 | 2.221 | 2.369 | 2.350 | 2.318 | 2.442 | 2.303 | 2.405 | 2.338 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 895   | 901   | 885   | 857   | 893   | 924   | 907   | 895   | 928   | 852   | 867   | 877   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 95    | 126   | 149   | 166   | 140   | 121   | 122   | 123   | 101   | 104   | 81    | 80    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.50  | 3.65  | 4.64  | 4.89  | 5.71  | 7.11  | 7.26  | 6.49  | 5.34  | 3.47  | 2.37  | 1.98  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.26  | 0.30  | 0.46  | 0.50  | 0.43  | 0.40  | 0.38  | 0.27  | 0.32  | 0.28  | 0.33  | 0.32  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)  | (e)                    | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | +0.64   | -0.45 | N/A   | N/A         | N/A     | +89     | +37     |

Nomenclature: See separate page